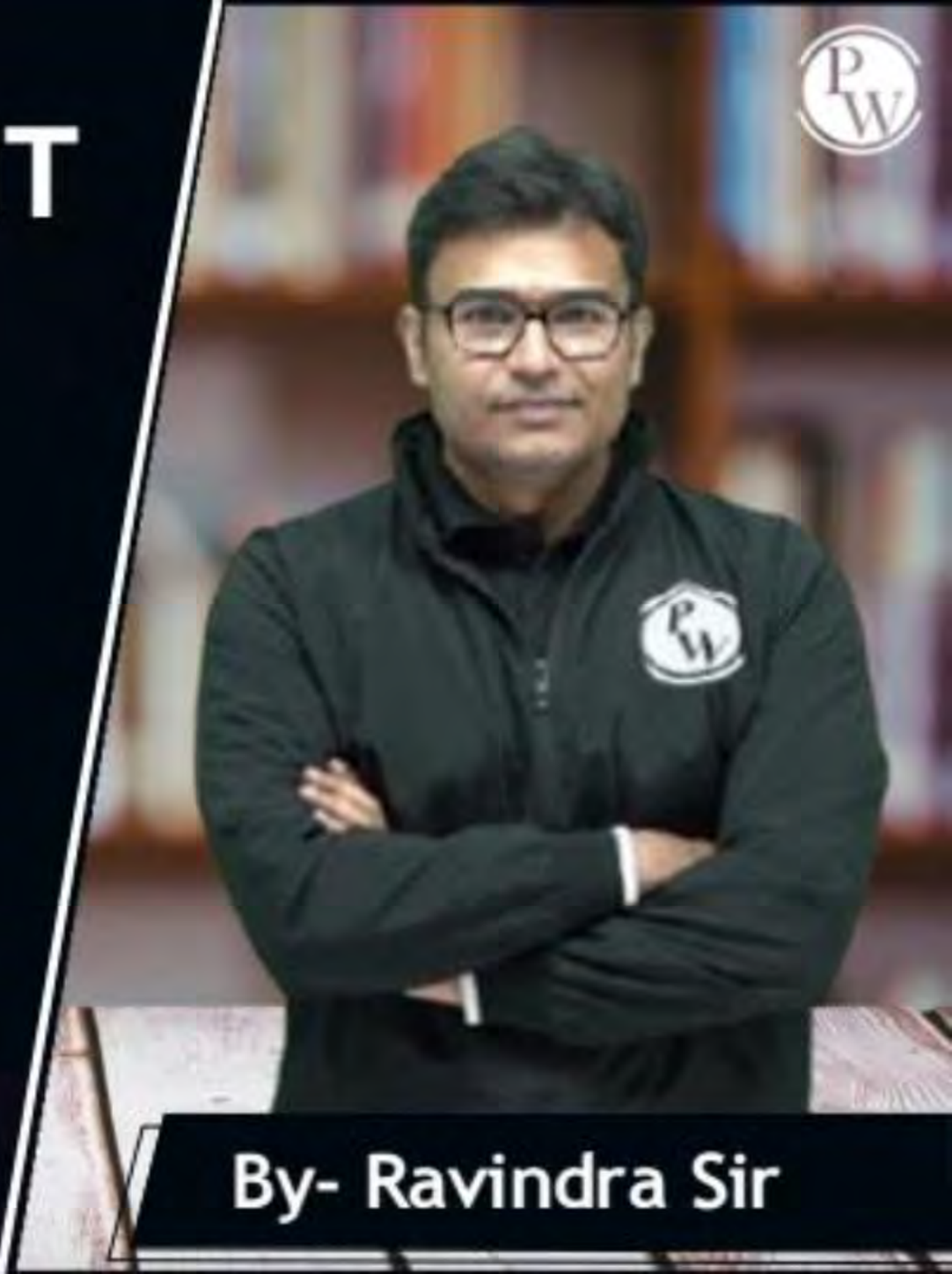
Computer Science & IT

COMPILER DESIGN
(CD)

Intermediate Code Generation

Lecture No. 1



By- Ravindra Sir

Recap of Previous Lecture



- Topic SOT
- Topic
- Topic

Topics to be Covered



- Topic
- Topic
- Topic

ICG

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Inspiring Stories : Mula Gabharu

Mula Gabharu – The Queen Who Raised Women Soldiers

Background: Mula Gabharu was the daughter of an Ahom king in Assam in the early 1500s.

Struggles: Her husband fell fighting an invading army. Her people were helpless and losing hope.

Achievements: She trained women warriors, led them into battle with courage, and confronted the enemy herself before being martyred.

Impact: If she could turn grief into action and lead women into battle, you can turn doubt into study and lead your rank up.



Inspiring Stories : B. N. Jagatap

B. N. Jagatap

Background: Grew up in a modest family in rural Maharashtra with no strong science background at home.

Education: PhD in Nuclear Magnetic Resonance (NMR) from IISc Bangalore.

Career Achievements: Improved NMR methods to study molecules in great detail, helping drug design and materials research.

Impact: Made high-end science tools more effective for Indian research labs and universities.



Inspiring Stories : Veeramangai Kuyili

Veeramangai Kuyili – The Woman Who Became a Human Torch

Background: Kuyili was a trusted soldier of queen Velu Nachiyar from Tamil Nadu in the 1700s.

Struggles: She worked without praise and lived in a time when women were rarely seen in battle.

Achievements: During one attack, she covered her body in ghee, set herself on fire, and jumped into the British armory. Her act became a key turning point in the war.

Impact: If she could light herself on fire to save her queen, you can light up that answer sheet with focus



Inspiring Stories : K. J. Rao

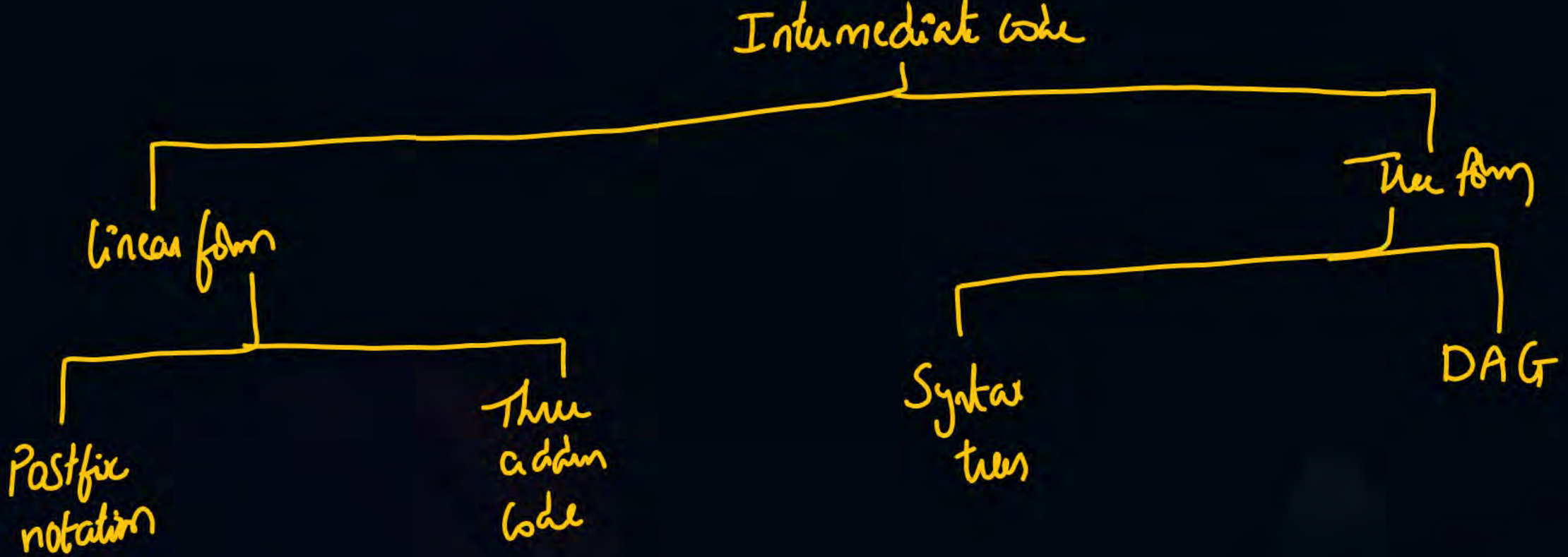
K. J. Rao

Background: Born in a small town in Karnataka with very limited means. Had to work hard to afford studies.

Education: PhD in Solid State Chemistry from IISc Bangalore.

Career Achievements: Developed new ceramic materials that can store energy and work in extreme heat.

Impact: Helped India make better energy storage systems and heat-resistant tools for industry.


Intermediate code
Linear form
Tree form
Postfix notation
Three address code
Syntax trees
DAG

$(a+b)*(a+b+c)$

Postfix:- $ab+ab+c+*$

* **Three add Code :-**

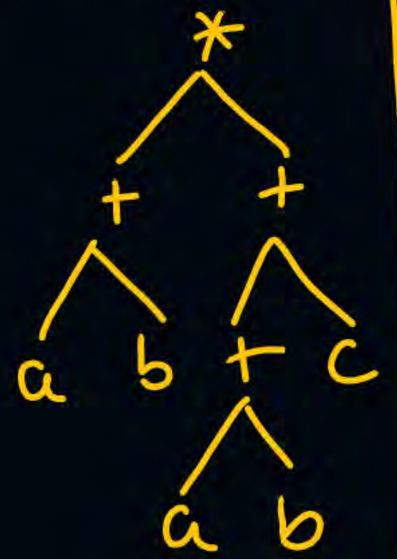
$t_1 = a+b$

$t_2 = a+b$

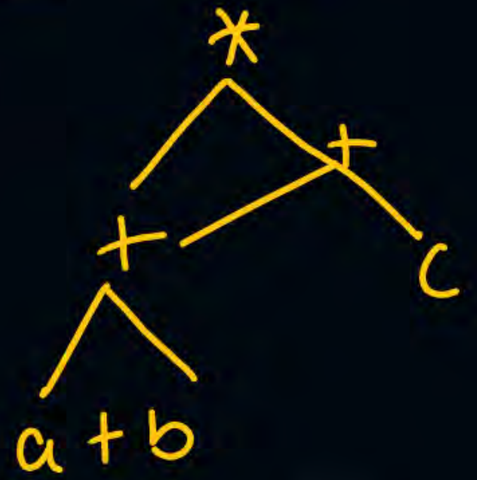
$t_3 = t_2+c$

$t_4 = t_1 * t_3$

Syntax tree :-



DAG :-



Types of 3 add Code :-

1) $x = y \text{ op } z$
2) $x = \text{op } z$
3) $x = y$
4) if x relop y goto L
5) goto L
6) $A[i] = x$
   $y = A[i]$
7) $x = *p$
   $y = \&x$

function calling

Param p_1
Param p_2
Param p_3
:
:
Param p_n
t = call f, n
return a

a(i, j)
Param i
Param j
t = call a, 2

$-(a+b)*(c+d)+(a+b+c)$

1) $t_1 = a+b$
2) $t_2 = -t_1$
3) $t_3 = c*d$
4) $t_4 = t_2 * t_3$
5) $t_5 = (a+b)$
6) $t_6 = t_5 + c$
7) $t_7 = t_4 + t_6$

3 add code

- Quadruples
- Triples
- Indirect triples

Quadruples :-

	opr	op1	op2	result
1)	+	a	b	t_1
2)	−	t_1	———	t_2
3)	+	c	d	t_3
4)	*	t_2	t_3	t_4
5)	+	a	b	t_5
6)	+	t_5	c	t_6
7)	+	t_4	t_6	t_7

adv :- Statements can be moved around

disadv :- Too much of space is wasted

Triplet

	op	op1	op2
1)	+	a	b
2)	-	(1)	
3)	+	c	d
4)	*	(2)	(3)
5)	+	a	b
6)	+	(5)	c
7)	+	(4)	(6)

Indirect triples :-

(i)	(1)
(ii)	(2)
(iii)	(3)
(iv)	(4)
(v)	(5)
(vi)	(6)
(vii)	(7)

Triplet

adv :- Space is not wasted

dis :- Statement cannot be moved

Indirect triples :-

adv :- Statements can be moved

dis :- Two memory accesses

→ if (a < b) then t = 1 else t = 0

Back patching

(i) if (a < b) goto (i+3)

(i+1) t = 0

(i+2) goto (i+4)

(i+3) t = 1

(i+4)

$a < b$ (t_1) and $c < d$ (t_2) & $e < f$ (t_3)

100) if ($a<b$) goto 103
101) $t_1 = 0$
102) goto 104
103) $t_1 = 1$
104) if ($c<d$) goto 107
105) $t_2 = 0$
106) goto 108
107) $t_2 = 1$

108) if ($e<f$) goto 111
109) $t_3 = 0$
110) goto 112
111) $t_3 = 1$
112) $t_4 = t_1$ and t_2
113) $t_5 = t_4$ & t_3

THANK - YOU